



Session-4 : Sensors & Perception

Submission Deadline: 11:59 PM, 25th June 2026

Submit drive link containing all the deliverables and a github repo of the modified files

Assignment 4: The Great Object Hunt

After weeks of development, your home assistant robot is finally ready for testing.

Well... almost.

The perception team has done their job. The robot can already recognize dozens of everyday objects using YOLOv8 and display them on screen. It can tell the difference between a bottle, a chair, a person, and many other household items.

But there is one problem.

The robot can see objects, yet it has absolutely no idea what to do with them.

If someone asks:

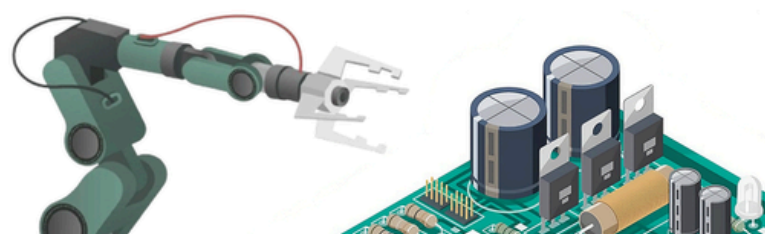
"Can you find my bottle?"

the robot simply stares at the screen and draws a bounding box around it. Not very helpful.

Your task is to transform this passive observer into an intelligent assistant capable of searching for a target, tracking it, navigating toward it, and stopping safely once it reaches its destination.

Welcome to **The Great Object Hunt !**

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What You Have Been Given

The starter perception package is what you made till resources of **session 4**.

The current YOLOv8 node can:

- Subscribe to camera images
- Run YOLO object detection
- Display bounding boxes and class labels
- Compute object center coordinates
- Show a dashboard containing detections and FPS

Your job is to build everything that comes after perception.

Remember:

Detecting an object is only the beginning.

A robot becomes useful when it can act on what it sees.

Stage 1 – Choose the Target

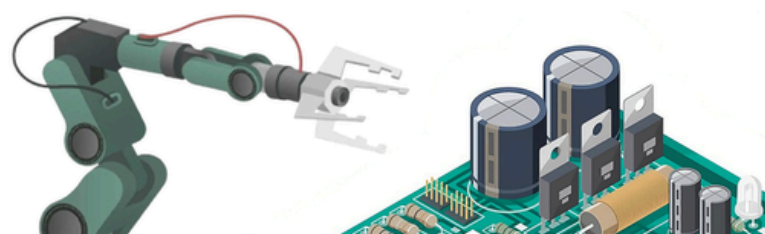
A user should be able to tell the robot what object they are looking for.

Examples:

- person
- fire hydrant
- chair
- bottle

Modify the system so that the robot accepts a target object (must be in the surrounding of robot, not very far) before beginning its mission.

Once a target has been selected, the robot should focus only on that object and ignore all other detections.





Expected Output:

Searching for: bottle

Hint:

- The starter code already provides the class name for every detected object.
- Think about where and how a user can provide input while the node is running.

Stage 2 — Teach the Robot Distance

The robot currently knows:

"There is a bottle in front of me."

But it does not know:

"The bottle is 2 meters away."

Subscribe to the depth camera and estimate the distance to the selected target.

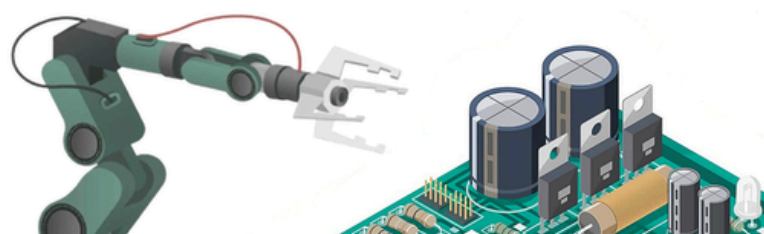
Expected Output:

Target Found

Distance to bottle: 2.35 m

Hint:

- You already have access to the center coordinates of each detected object.
- Can those coordinates help you determine where to read information from the depth image?





Stage 3 — The Search Begins

What if the target is not visible?

Imagine being asked to find a bottle and simply standing still forever.

A useful robot should actively search for its target.

Implement a search behaviour that allows the robot to scan its surroundings until the requested object appears.

Expected Behaviour:

Searching...

Searching...

Searching...

Target Found!

Hint:

- How would you start searching a room for a missing object?
- Think about simple robot motions that increase the camera's field of view.

Stage 5 — Begin the Hunt

The target has been found.

The robot is facing it.

Now it must approach it.

Combine:

- Object Detection
- Distance Estimation
- Target Tracking
- Robot Motion

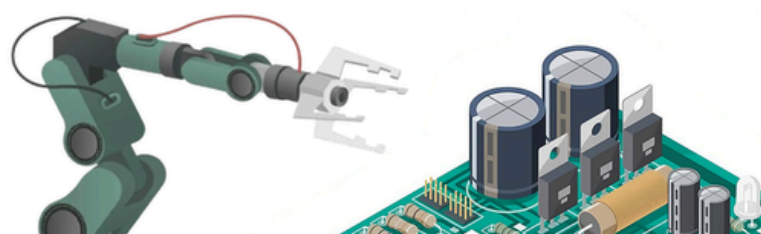
to create a fully autonomous object-seeking behaviour.

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The robot should continuously update its decisions as new camera frames arrive.

Expected Behaviour:

Target Locked

Distance: 2.4 m

Distance: 1.8 m

Distance: 1.1 m

Distance: 0.7 m. . .

Hint:

- The starter code can already detect objects.
- The challenge now is deciding how the robot should move based on what it sees.

Stage 6 – Mission Completion

No one wants a robot crashing into furniture.

Choose a safe stopping distance for your robot.

When the target is reached:

- Stop all robot motion
- Publish zero velocity
- End the mission

Expected Output:

Mission Completed

Target Reached Successfully

Hint:

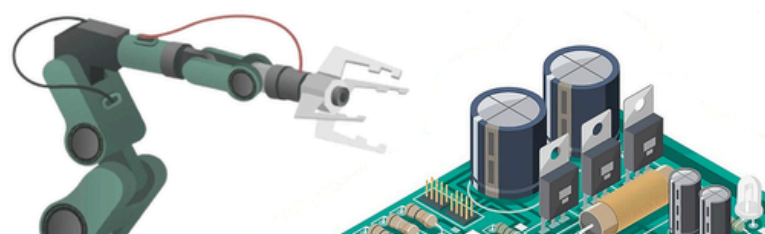
- How can the robot determine that it is close enough to the object?

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Bonus Challenges (Not compulsory)

Bonus 1 — Smarter Distance Estimation

Depth values are not always perfect.
Sometimes they may fluctuate or contain invalid measurements.

Can you improve the stability of your distance estimate?

Hints:

- Is a single depth pixel always reliable?
- What happens if that pixel contains noise?
- Try examining a small region around the object's center.
- Statistical techniques such as averaging or median filtering may help.

Bonus 2 — Mission Control Dashboard

Upgrade the existing dashboard.

Display useful information such as:

TARGET : bottle

STATUS : TRACKING

DISTANCE : 1.24 m

MODE : APPROACH

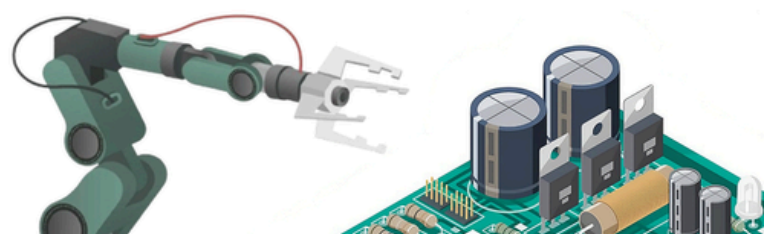
A good dashboard helps operators understand what the robot is thinking.

Hints:

- The dashboard is already being drawn every frame.
- Consider what information would be useful while debugging the robot.
- Try updating dashboard values in real time as the robot changes state.

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Bonus 3 — Continuous Assistant Mode

After completing one mission:

Mission Completed !

the robot should immediately be ready for another.

Example:

(after one successful mission it should ask for another)

Enter target object:

The user should not need to restart the program.

Hints:

- Which variables should be reset after a successful mission?
- Can the target input system continue running throughout the program?
- Think about how a real home assistant would behave after completing a task.

Deliverables

Submit:

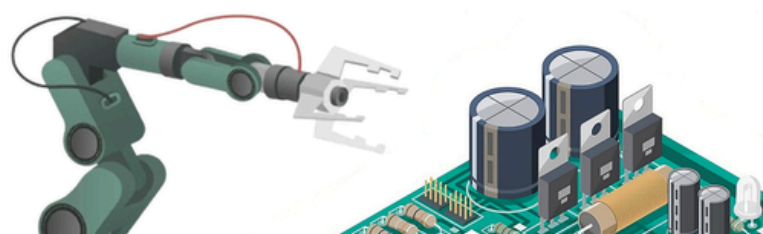
- **Source Code**

Include all modifications made to the starter package.

- **Demonstration Video**

Your video should clearly demonstrate:

- Target selection
- Search behaviour
- Target detection
- Distance estimation
- Target tracking
- Autonomous navigation
- Mission completion





• Short Report

Briefly explain:

- How you estimated distance ?
- How your search behaviour works ?
- How your robot decides when to move and stop ?

Final Objective

By the end of this assignment, your robot should be capable of:

SEARCH → DETECT → TRACK → APPROACH → COMPLETE

A camera can recognize objects.

An intelligent robot can find them.

Good luck, and happy hunting!

